

III Year

Roll No.

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Course Name: Food and Dairy Microbiology
Course Code: 303

Branch(s): Microbiology

CO-I: Important microorganisms in food (meat, poultry, vegetables, dairy products and dehydrated foods) — moulds, yeasts, yeast-like fungi and bacteria.

CO-II: Principles of food preservation and spoilage of foods.

Time: 120 Min

Max. Marks: 60

Instructions:

1. Attempt all the sections.
2. Assume missing data suitably, if any.

KL: Bloom's knowledge Level (K1, K2, K3, K4, K5, K6)

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate, K6-Create

Attempt all questions of this section.

Section-A

Q.No.

Questions		(2 x 6 = 12 Marks)	
		CO	KL
1.	a Name any two important bacteria associated with meat spoilage.	1	K1
	b Define yeast-like fungi and give one example.	1	K1
	c Mention any two important moulds involved in spoilage of vegetables.	1	K2
	d Define food preservation.	2	K1
	e What is food spoilage?	2	K2
	f Differentiate between homofermentative and heterofermentative LAB with respect to their major fermentation products.	2	K3

Attempt any two questions of this section.

Section-B

Q.No.

Questions		(7 x 2 = 14 Marks)	
		CO	KL
2.	Explain the important microorganisms associated with meat, including major bacteria and moulds responsible for spoilage.	1	K1
3.	Describe the important microorganisms associated with dairy products. Explain their role in desirable fermentation.	1	K1
4.	Explain the microbial flora of vegetables and discuss the major microorganisms responsible for vegetable spoilage.	1	K1

Section-C

Attempt any two questions of this section.

(7 x 2 = 14 Marks)

5.	Explain the principles of food preservation. Discuss how control of temperature, moisture and pH inhibits microbial growth.	2	K3
6.	Describe the different types of food spoilage and explain the major factors affecting microbial spoilage of foods.	2	K2

7.	Explain the microorganisms important in the spoilage of dehydrated foods with special reference to moulds, yeasts and bacteria.	2	K2
		2	

Attempt all questions of this section. Marks)		<u>Section-D</u> (10 x 2 = 20)	
Q.No.	Questions	CO	KL
8.	Explain the microbial spoilage of dairy products and the resulting changes in flavour, odour, texture and appearance.	1	K2
	OR		
9.	Explain the important microorganisms causing spoilage of meat and poultry and the resulting spoilage changes.	1	K2
	Explain physical and chemical methods of food preservation with examples.	2	K3
	OR		
	Explain the mechanism of microbial food spoilage. Discuss the major intrinsic and extrinsic factors that influence the growth of microorganisms in foods.	2	K3